Multimodal Artificial Intelligence for Accessible Picture-Book Narration

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***Abstract*—This paper presents a multimodal artificial intelligence (AI) pipeline that generates accessible narrations of children's picture books for blind and low-vision readers. The system extracts printed text from page images, generates grounded visual descriptions of illustrations, maintains story context across pages, and produces concise accessible narration that preserves the original author's text while adding relevant visual information. A post-processing pipeline removes unsupported sensory language, enforces word-budget limits, protects dialogue, and maintains character identity across pages. Evaluation combined automated quality metrics and a small within-subjects user preference study with 16 adult blind and low-vision participants, all of whom self-reported a visual impairment. Across 512 preference judgments, participants selected the visually enriched narration in 76.6% of all responses, including neutral responses. Excluding neutral responses, the enriched narration accounted for 87.7% of sided votes, with the strongest preference for description quality at 94.2% of sided votes. These results support the feasibility of using multimodal AI to generate accessible first-draft narrations for illustrated children's books, while child-reader evaluation remains future work.**

Keywords—accessible reading; picture books; multimodal AI; visual impairment; narrative generation; blind and low-vision

I. Introduction

Children's picture books are among a child's earliest encounters with narrative, introducing sequencing, character identity, emotion, cause and effect, and resolution. For sighted children, much of this meaning is carried not by the printed words but by the illustrations: facial expressions, body language, setting, and visual cues that determine how a story is understood. For blind and low-vision children, this visual layer is frequently inaccessible.

The accessibility gap is substantial: the World Health Organization estimates at least 2.2 billion people worldwide have near or distance vision impairment [1]. Screen readers and optical character recognition (OCR) recover printed text but do not explain what an illustration contributes to a story, and professionally produced audiobooks are costly and unavailable for many titles.

We present a multimodal artificial intelligence (AI) pipeline for accessible picture-book narration. Given a page, the system extracts printed text, analyzes the illustration, generates a concise visual description, and integrates it into the original narrative. Rather than replacing the author's text or producing a generic caption, it preserves the original text while adding the visual context a blind or low-vision child needs to understand the scene.

This work makes three contributions. First, it frames picture-book accessibility as a narrative-integration problem rather than image captioning. Second, it presents a multimodal pipeline that combines text extraction, illustration analysis, continuity tracking, constrained generation, and post-processing safeguards. Third, it proposes evaluation criteria tailored to accessible storytelling: unsupported visual claims, plot-continuity errors, and preservation of original text.

II. Background and Related Work

A. Accessible Reading Technologies

Screen readers (JAWS, NVDA, VoiceOver) and OCR have substantially improved access to textual information for blind and low-vision readers, but are effective only when meaning resides primarily in text.

Picture books are harder: the story is distributed across text and image, and a page may carry only a short sentence while the illustration supplies setting, emotion, and plot. The DAISY format offers navigable reading for print-disabled users [2] but does not describe illustrations, and tactile or human-adapted books require manual effort that does not scale.

B. Image Captioning and Visual Description

Vision-language models have advanced image description: CLIP learns transferable visual representations [3], BLIP unifies vision-language understanding and generation [4], and LLaVA connects visual encoders to large language models [5].

These provide building blocks, but conventional captioning is insufficient for picture books. A caption such as "a child stands next to a dog" may be accurate yet narratively incomplete; it does not convey whether the child is frightened or why the scene matters. The task is closer to narrative augmentation than captioning.

C. Automatic Audio Description and Alt-Text Generation

Automatic audio description (AD) describes visual information in video, such as actions, expressions, and scene changes, so blind viewers can follow visual content [8], [9]. It

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is closely related to our goal of translating visual information into language for nonvisual access.

Automatic alt-text generation produces textual descriptions of images for screen-reader users, and prior work shows users need descriptions reflecting task, context, and detail rather than object labels alone [10], [11]. However, alt text replaces a single image, whereas picture-book narration must integrate visual information into existing literary text while preserving wording, pacing, dialogue, and cross-page continuity.

Work on tactile and accessible picture books shows illustrations matter for young blind readers, supporting enjoyment and early literacy [12]; existing tactile, braille, and human-described editions, however, require manual adaptation and do not scale.

D. Multimodal Language Models for Narrative Augmentation

Multimodal language models enable accessible storytelling by combining visual interpretation with fluent generation, but they also introduce risks: hallucinated details, inferred unsupported emotions, paraphrased original text, and lost character identity across pages. These risks matter especially because blind readers cannot independently verify visual content. Our pipeline addresses them by separating visual evidence extraction from narrative generation, constraining added text, preserving dialogue, and applying post-processing safeguards.

E. Gap Addressed by This Work

Prior tools for text access, accessible publishing, captioning, alt text, and audio description [2]–[12] do not fully address illustrated narrative access in children's books. OCR misses illustrations, alt text does not preserve literary pacing, audio description targets time-based media rather than page-based narratives, and tactile books require manual production. We frame the problem as faithful narrative integration: preserving the author's text while inserting concise, grounded visual descriptions that maintain character identity, dialogue integrity, and cross-page continuity.

TABLE I. COMPARISON OF EXISTING ACCESSIBILITY APPROACHES ACROSS CAPABILITIES REQUIRED FOR ACCESSIBLE PICTURE-BOOK NARRATION

APPROACH	READS TEXT	DESCRIBES ILLUSTRATIONS	INTEGRATES WITH NARRATIVE	PRESERVES DIALOGUE	CROSS-PAGE CONTINUITY
Bookshare / DAISY	✓	Limited / manual	✗	✓	✓
Alt text / image description	✗	✓	✗	✗	✗
Automatic audio description	✗	✓	✗	✗	Limited
Proposed multimodal narration	✓	✓	✓	✓	✓

III. SYSTEM ARCHITECTURE

The system is a multi-stage pipeline that converts picture-book pages into accessible narrative text, built around a separation between visual-evidence extraction and narrative generation.

Each page passes through five stages: image encoding, page classification, text extraction, visual-description generation, and accessible-narrative generation. For complete books, a shared story-context object preserves character identity, limits repetition, and tracks recent events (Fig. 1).

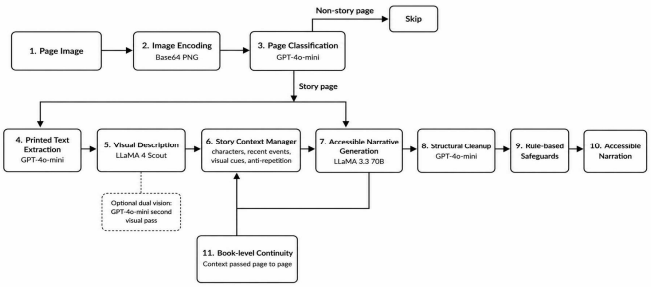


Fig. 1. System architecture. Picture book page image → image encoding → page classification (GPT-4o-mini) → printed text extraction (GPT-4o-mini) → visual description (LLaMA 4 Scout) → story context memory → constrained narrative generation (LLaMA 3.3 70B) → GPT-4o-mini structural cleanup → rule-based post-processing → accessible narration.

GPT-4o-mini performs structured extraction, LLaMA-4-Scout generates grounded visual descriptions, and LLaMA-3.3-70B performs narration. Model selection was based on architectural fit, reliability, and cost. Fixed versions were used throughout.

For reproducibility, all inference used fixed model versions (gpt-4o-mini-2024-07-18, meta-llama/llama-4-scout-17b-16e-instruct, and llama-3.3-70b-versatile) via the OpenAI and Groq APIs during May 2026.

A. Page Representation and Image Encoding

Each page is an independent visual input within an ordered book sequence. It is converted to a base64 Portable Network Graphics image and passed to the vision model; the same image serves both text extraction and visual description, separating OCR-like recognition from scene interpretation. Deterministic settings (fixed seed, low temperature) reduce output variability.

B. Printed Text Extraction

GPT-4o-mini extracts readable printed text (dialogue, captions, signs, labels), instructed to return only readable text, preserve line breaks, avoid summarization, and return NO_TEXT when none is detected. A guardrail rejects unusually long single-line outputs that may indicate hallucination. The output, extracted_text, is the textual anchor for narration.

C. Visual Description Generation

LLaMA-4-Scout generates a concise visual description of the illustration, restricted to observable details only: expressions, positioning, objects, salient clothing, background, colors, and spatial relationships. Additions are limited proportionally to page length: at most 50% of the original word count, with a floor of eight and a ceiling of eighteen words, consistent with picture-book sentence norms.

To reduce hallucination, the prompt discourages fabricated sensory or emotional content (e.g., "sounds of nature," "air feels

warm"). Detected critical cues, such as tears, open mouths, wide eyes, fallen objects, background, colors, sizes, and spatial relations, are stored in the story context for later stages.

D. Story Context Memory

The StoryContextManager stores cross-page state: page index, character identities and confidence, recent events, prior context, and critical visual details. Identity tracking maps character names to roles or species via pattern matching, then injects them as hard constraints (e.g., "CHARACTER IDENTITIES—NEVER MIX UP") to prevent confusion of recurring characters.

It tracks previously described details and recent events to reduce repetition while preserving continuity.

E. Accessible Narrative Generation

LLaMA-3.3-70B generates the final narration from the extracted text, visual description, identity constraints, anti-repetition reminders, recent context, and critical cues. When text exists, the model begins with the exact story text and appends factual visual details; otherwise it narrates from the description alone. The medium-detail elementary setting targets about 70 to 90 words on the first page and shorter continuations thereafter.

F. Post-Generation Safeguards

Two post-processing layers follow. A GPT-4o-mini structural-cleanup pass corrects common LLaMA failures, including reordered sentences, semantic repetition, descriptions placed inside dialogue quotes, and altered original wording; if the cleaned output is over-shortened, the system falls back to the original extracted text. A rule-based pass then removes meta-commentary and unsupported emotional language, converts subjective interpretations to observable cues, and normalizes punctuation.

IV. Post-Processing Pipeline

Raw large language model output requires post-processing to improve factuality, preserve original text, remove unsupported language, and reduce repetition.

A. Output Normalization and Meta-Text Removal

Normalization removes meta-commentary, stray whitespace, and punctuation artifacts; lines beginning with phrases such as "let me" are filtered so the output reads as a story rather than an AI response.

B. Unsupported Sensory and Emotional Language Filtering

Unsupported sensory and emotional phrases (e.g., "sounds of nature," "joyful atmosphere") are removed, and subjective interpretations are converted to observable cues ("tearful expression" becomes "with tears"), enforcing description before inference.

C. Original Text Preservation

A spine-reconstruction step locates all original sentences in the output, reorders displaced ones, and restores absent or paraphrased sentences verbatim, retaining novel sentences adjacent to their context. This yields OTPR = 1.00 across all evaluated pages (Table III).

D. Repetition Control

Repeated machine-generated descriptions distract when read aloud, so the context manager issues anti-repetition reminders for previously described details, and sentence-level cleanup removes near-duplicate phrases.

E. Character Identity Safeguards

To preserve identity, extracted identities are injected as pre-generation constraints and re-checked in post-processing; once established, characters are referenced by name rather than reintroduced by type.

The system constrains both the intermediate visual description and the final narration to preserve picture-book rhythm and brevity. The visual description stage enforces approximate word caps based on content level. For the final narration, a proportional word budget is applied: at most $\min(40, \max(15, \lfloor \text{original_word_count} \times 0.5 \rfloor))$ words may be added above the original text, with a hard post-processing trim at sentence boundaries if the model exceeds this budget. The proportional design ensures additions scale with page density while staying within one to two picture-book sentences for typical short pages; tighter calibration of the floor and ceiling toward genre norms is noted as future work.

F. Dialogue Protection

Dialogue carries character voice and rhythm, so on dialogue-heavy pages the system limits insertions to one sentence placed before or after the complete speech block; a quote-depth tracker defers insertions until multi-sentence dialogue closes.

G. Summary of Post-Processing Steps

TABLE II. Post-Processing Steps for Accessible Narration

Step	Function	Failure Mode Addressed
Output normalization	Removes extra whitespace, punctuation artifacts, and meta-commentary	AI-style explanation in final narration
Unsupported language filtering	Removes fabricated sensory and emotional phrases	Hallucinated sounds, feelings, or atmosphere
Factual replacement	Converts subjective interpretations into observable cues	Inference-heavy descriptions such as “worried”
Original text preservation	Keeps extracted story text as the narrative spine	Paraphrasing or reordering of author text
Repetition control	Tracks prior clothing, positioning, and visual details	Repetitive narration across pages
Character identity safeguard	Maintains named character identities across pages	Character confusion or repeated re-identification
Length control	Applies word targets and sentence-boundary truncation	Overly long visual insertions
Dialogue protection	Keeps visual insertions outside quoted speech	Broken or confusing dialogue
GPT-4o-mini structural cleanup	Detects and corrects structural errors in LLaMA output: reordered sentences, semantic repetition, dialogue violations, altered original text	LLaMA reordering or paraphrasing original sentences; inserting descriptions inside multi-sentence dialogue; altering original book text
Safety fallback	Returns original text when generation appears malformed	Truncated, empty, or unreliable output

Table II summarizes the main post-processing functions and the failure modes they address.

H. Computational Cost

Processing cost is ~$0.004 per page in API fees (Groq and OpenAI), under $0.10 per book, making the pipeline inexpensive to deploy.

V. Evaluation

Standard captioning metrics are insufficient here, so the evaluation combines automated quality checks, a qualitative example, and a user preference study.

A. Automated Quality Metrics

Four page-level metrics assess primary risks. UCR (Unsupported Claim Rate) is the fraction of sentences with visual claims absent from the image description; PCER (Plot Continuity Error Rate) is the fraction of cross-page checkpoints introducing a contradicting state; DPR (Dialogue Preservation Rate) and OTPR (Original Text Preservation Rate) are the fractions of original dialogue and text tokens preserved. DPR and OTPR use string overlap; UCR and PCER use GPT-4o-mini as the judge.

Metrics were computed across all four study books (Terrance Turtle's New Home, The Naughty Monkey, Captain Fantastic, and Homework Yuck), totaling 61 pages. The judge flagged 1.3% UCR (Homework Yuck) and 20.0% PCER (Captain Fantastic); manual review found both to be false positives, caused by simultaneous actions and by a check applied to source rather than output, with zero genuine violations across all 61 pages (Table III).

TABLE III. Automated Quality Metrics Across Four Books (61 Pages)

Book	Pages	UCR (auto)	UCR (human)	PCER (auto)	PCER (human)	DPR	OTPR	WER
Terrance Turtle's New Home	12	0.0%	0.0%	0.0%	0.0%	1.00	1.00	1.30x
The Naughty Monkey	19	0.0%	0.0%	0.0%	0.0%	1.00	1.00	1.31x
Captain Fantastic	10	0.0%	0.0%	20.0%	0.0%	1.00	1.00	1.22x
Homework Yuck	20	1.3%	0.0%	0.0%	0.0%	1.00	1.00	1.51x
All books	**61**	**0.3%**	**0.0%**	**5.0%**	**0.0%**	**1.00**	**1.00**	**1.36x**

UCR = Unsupported Claim Rate; PCER = Plot Continuity Error Rate; DPR = Dialogue Preservation Rate; OTPR = Original Text Preservation Rate; WER = Word Expansion Ratio. Validation through manual review.

Across all four books the system preserves text and dialogue perfectly (DPR = OTPR = 1.00) while adding measured visual context (mean WER 1.36). The automated flags were not confirmed by human review, indicating scorer over-flagging rather than genuine hallucination; scorer refinement is future work. UCR remains the most critical category to monitor.

B. Qualitative Example

Figure 2 illustrates grounded environmental and character-description enrichment while preserving dialogue verbatim and in order.

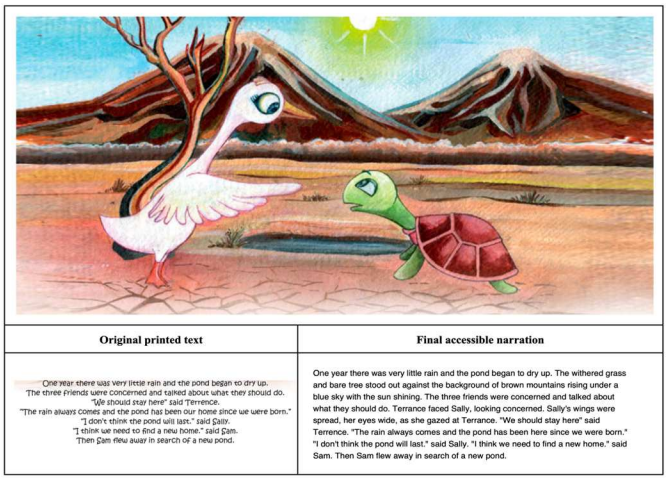


Fig. 2. Representative evaluation example. Top: original picture-book illustration. Bottom left: corresponding original book text. Bottom right: final accessible narration generated by the proposed system, preserving the original story while integrating grounded visual information from the illustration. The original text is shown only for comparison during evaluation and is not presented separately to readers.

C. User Preference Study

A within-subjects preference study tested whether blind and low-vision readers preferred original or enriched narration; the design controls for individual variation in reading style, impairment type, and language background that could dominate a small between-group sample. Eight excerpts from three books (Terrance Turtle's New Home, Ashley the Rabbit and Baba, Captain Fantastic) were compared on four dimensions: better descriptions, more enjoyable, easier to remember, and recommend to others.

1) Adaptive Data Collection

A three-participant pilot revealed that Google Forms' default required-field styling (white-on-red) was illegible under some low-vision conditions, causing misidentified options; all pilot responses were excluded to avoid test-retest effects. The final study used an accessibility-corrected, counterbalanced instrument with 16 participants, randomizing version order per excerpt and re-coding all responses so Version 1 always denotes original and Version 2 enriched. This yields a methodological point: accessible survey instruments must themselves be accessibility-audited, since standard styling can introduce bias invisible in aggregate data.

The study was reviewed and approved by Vista Center for the Blind and Low Vision (Palo Alto, CA) under its organizational research guidelines. All participants gave voluntary informed consent and could stop at any time; no direct identifiers were collected for analysis, demographic data were analyzed only in aggregate, and compensation was administered separately from the research dataset.

2) Participants

The final dataset comprised 16 participants and 512 judgments (8 excerpts × 4 questions × 16). All 16 self-reported a visual impairment (VI): five blind from birth, six acquired, five low vision. Ages ranged widely (two 18–25, three 25–35, two 35–45, one 45–55, five 55–65, three 65+); twelve were native English speakers. Access methods (multiple allowed):

screen readers (11), braille (6), large print (6), text-to-speech (TTS) (1). All were recruited via Vista Center; each received $20. All were adults; pediatric evaluation is planned.

3) Statistical Analysis

Excluding neutral responses left 447 sided votes across 16 participants. The primary analysis is a participant-level sign test: per participant, the share of sided votes for enriched was computed and a two-sided binomial test applied to the count preferring enriched ($k = 15$, $n = 16$, $p = 0.000519$), accounting for within-participant clustering. For context, 87.7% of sided votes favored enriched. Gwet's first-order agreement coefficient (AC1) (participants as raters, 32 items) measured agreement; Fleiss' κ is inappropriate here due to the prevalence paradox [6] ($\kappa = 0.017$ vs. AC1 = 0.530). Significance threshold $\alpha = 0.05$.

4) Results

Of 512 judgments, 392 chose enriched (76.6%), 55 original (10.7%), and 65 neutral. Among 447 sided votes, 87.7% favored enriched (95% confidence interval (CI) [84.3%, 90.4%]; odds ratio 7.1:1). At the participant level, 15 of 16 cast a majority for enriched (93.8%, CI [69.8%, 99.8%]; sign test $p = 0.000519$). Gwet's AC1 = 0.530. Enriched was preferred in all 8 excerpts and all 4 dimensions (Tables IV, V).

The strongest preference was for description quality (114/121 sided, 94.2%; AC1 = 0.780), which directly supports the core premise. The weakest, though still significant, was memorability (79/104, 76.0%; AC1 = 0.225), consistent with cognitive-load research: enrichment improves in-the-moment comprehension more than recall, which supports a restrained, minimal-addition design.

TABLE IV. USER PREFERENCE RESULTS BY QUESTION TYPE (SIDED % = V2 OF V1+V2 ONLY)

PREFERENCE QUESTION	V1	V2	NEUTRAL	SIDED V2%	95% CI
Better descriptions	7	114	7	94.2%	[88.5%–97.2%]
More enjoyable	15	97	16	86.6%	[79.1%–91.7%]
Easier to remember	25	79	24	76.0%	[66.9%–83.2%]
Recommend for readers	8	102	18	92.7%	[86.3%–96.3%]
Total	**55**	**392**	**65**	**87.7%**	**[84.3%–90.4%]**

Vote-level proportions with 95% Wilson CIs. Primary significance established via participant-level sign test (k = 15/16, p = 0.000519), reported in text.

TABLE V. VERSION 2 PREFERENCE BY EXCERPT

EXCERPT	VERSION 2 SELECTIONS	TOTAL JUDGMENTS	V2 SHARE
1	53	64	82.8%
2	54	64	84.4%
3	45	64	70.3%
4	41	64	64.1%
5	52	64	81.2%
6	53	64	82.8%
7	47	64	73.4%
8	47	64	73.4%

The enriched version was preferred for every one of the eight excerpts, with sided preference rates ranging from 78.9% to 93.0% (all $p < 0.001$). This consistency across excerpts, spanning three different books, different genres (animal fable, comedy, science fiction), and different narrative structures, indicates the preference is not driven by a single unusually strong example.

TABLE VI. STRATIFIED PREFERENCE RATES: ENRICHED VERSION (SIDED VOTES)

SUBGROUP	*N* (PARTICIPANTS)	ENRICHED V2 %	95% CI
VI Type: Blind from birth	5	88.7%	[82.2%–92.8%]
VI Type: Acquired	6	89.3%	[83.7%–93.1%]
VI Type: Low vision	5	85.0%	[78.2%–90.0%]
Age: 18–35	5	83.5%	[76.6%–88.8%]
Age: 35–55	3	74.0%	[63.5%–82.0%]
Age: 55 and above	8	93.5%	[89.2%–95.9%]
Access: Screen reader	11	94.3%	[91.3%–96.5%]
Access: Braille	6	87.1%	[81.0%–91.2%]
Access: Large print	6	84.6%	[78.3%–89.2%]
Access: TTS	1	23.8%	[12.7%–43.4%]
Source: Vista Center	12	91.1%	[87.5%–93.7%]
Source: Facebook VI community	4	77.3%	[69.1%–84.4%]
Native English: Yes	12	87.5%	[83.5%–90.6%]
Native English: No	4	88.3%	[81.1%–93.1%]

Vote-level proportions with 95% Wilson CIs. Subgroup sizes are too small for independent participant-level significance claims.

The atypical result for the single TTS participant motivates future evaluation of access-method-specific narration settings. The consistency seen across every other subgroup suggests the preference reflects the value of visual enrichment rather than any single demographic.

VI. DISCUSSION

Preferences consistently favored enriched narration across excerpts, evaluation dimensions, impairment types, recruitment sources, and age groups, with the strongest effect for description quality (94.2%, AC1=0.780).

The memorability result (76.0%, AC1 = 0.225) is the most informative for design. It was the weakest dimension, which is consistent with cognitive-load theory: added detail aids immediate comprehension but can burden recall. This motivates the word-budget and age-based length settings, which add the minimum context needed rather than the maximum possible.

The single TTS user, who preferred the original in 76.2% of sided votes, points to an access-method dependency. Screen-reader and braille users can navigate and re-read, so an added sentence costs little, whereas linear TTS makes each addition an interruption. A future TTS-optimized mode could shorten additions, place them at paragraph boundaries, or turn them off.

Although the system targets children, this evaluation deliberately studied adult VI readers, who face the same gap and choose accessible materials for children in their care; their strong preference validates the approach at the content level. In optional anonymous comments collected by Vista Center as part of the post-survey feedback, several participants who identified themselves as parents or grandparents described using the enriched narrations while reading with children and reported positive experiences. Other participants noted that they use children's picture books for English-language learning.

These qualitative observations suggest potential real-world utility beyond the formal study outcomes but do not substitute for direct evaluation with children. Whether young children benefit equally is open, and pediatric evaluation is the next phase.

The data-collection process also contributes methodologically: the inaccessible form styling produced bias detectable only through participant follow-up, showing that survey instruments in VI research need the same accessibility audit as the technology under study.

Reliability rests less on any single model call than on the surrounding controls: identity tracking, anti-repetition, word budgets, dialogue protection, and conservative language. The system is best understood as an assistive first-draft generator, not a replacement for human review in educational or publishing use.

VII. Conclusion

We presented a multimodal AI pipeline that enriches picture-book narration while preserving the original story.

Across automated metrics, a qualitative example, and the preference study, participants preferred the enriched version on all eight passages and four dimensions; excluding neutral votes, enriched accounted for 87.7% of sided votes, strongest for description quality (94.2%). Visually integrated narration can improve reading for blind and low-vision readers when additions stay concise and story-relevant.

The evaluation has limits: it is small, preference-based, and does not yet measure comprehension through controlled listening tasks. Additionally, the sample of 16 participants limits statistical power and the generalizability of subgroup findings; therefore, the results should be interpreted as evidence of feasibility and preference rather than population-level effectiveness. The findings may not generalize to wordless books, highly stylized artwork, dense visual scenes, comics, culturally specific imagery, or books with nonlinear page layouts. Hallucination and continuity assessments relied on manual verification following automated screening. Future work will include independent human evaluation to further strengthen reliability, scaling to full-length books, running larger studies, adding comprehension-based evaluation, and evaluating performance across a wider range of illustration styles and book formats. Future evaluation should compare the proposed system against generic captioning, standalone alt text, human-authored descriptions, and alternative multimodal models using both preference and comprehension measures.

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